

Coffee Kiosk POS - Customer Frontend

Test Estimation Report

Document field	Value
Project	Coffee Kiosk POS
Module	Customer - Frontend
Reference	Coffee Kiosk POS Customer Frontend FRS v1.0
Prepared by	Kushagra Sinha (QA Engineer)
Version	1.0
Date	25 September 2026
Status	Provisional - pending FRS approval

Table of Contents

- [1. Purpose](#)
- [2. Estimation Basis](#)
 - [Estimation Technique](#)
- [3. Testing Scope](#)
 - [In Scope](#)
 - [Out of Scope](#)
- [4. Assumptions and Dependencies](#)
 - [Assumptions](#)
 - [Dependencies](#)
- [5. WBS Estimation Matrix](#)
- [6. Effort and Duration Summary](#)
 - [Indicative Schedule](#)
- [7. Resource Requirements](#)
 - [Human Resources](#)
 - [Environment and Tools](#)
- [8. Risks and Mitigation](#)
- [9. Cost and Change Control](#)
- [10. Approval](#)

The referenced FRS is currently marked "Draft for review" with sign-off pending. This estimate becomes the planning baseline after FRS approval. Material scope changes require re-estimation.

1. Purpose

This report estimates the manual QA effort, duration, resources, dependencies, and risks required to test the Coffee Kiosk POS customer-facing frontend. It is intended to support project planning before detailed test cases are prepared.

2. Estimation Basis

The estimate covers all 48 requirements in FRS v1.0:

Requirement group	Count
Food Catalog	7
Cart	6
Checkout	8
Payment Gateway	7
Order Confirmation and Receipt	5
Non-functional requirements	15
Total	48

Estimation technique

- **Work Breakdown Structure (WBS):** Used to estimate each QA activity separately.
- **Expert judgement:** Used to estimate design and execution effort based on feature complexity.
- **Three-point estimation:** Applied to uncertain work, especially payment testing and defect retesting.

Three-point formula:

$$\text{Estimated effort} = (\text{Optimistic} + 4 \times \text{Most Likely} + \text{Pessimistic}) \div 6$$

Example for payment execution: $(6 + 4 \times 8 + 12) \div 6 = 8.3$ hours, rounded to 8 planning hours.

3. Testing Scope

In scope

- Manual functional testing of Food Catalog, Cart, Checkout, Payment, and Receipt features.
- Positive, negative, boundary, navigation, and error-handling scenarios.
- Customer-visible UPI and Card payment integration behaviour.
- Coupon validation, totals, taxes, and discount calculations.
- Touchscreen UI, usability, and basic accessibility checks.
- Customer-session timeout, privacy, and protected-role access checks.
- Supported Chromium-based kiosk browser and approved landscape resolution.
- Defect reporting, retesting, regression testing, RTM, and test summary reporting.

Out of scope

- Brew Crew and IT Admin interfaces.
- Direct API, database, and AWS service testing.
- Internal testing of the third-party payment provider.
- Load, stress, and full performance testing.
- Penetration testing or formal security certification.

- Test automation.
- KDS, thermal printer, cash payment, or physical card-terminal testing.

4. Assumptions and Dependencies

Assumptions

- FRS v1.0 is approved without major functional changes.
- One manual QA Engineer is assigned full-time, supported by a QA Lead for review.
- One person-day equals 7 productive QA hours.
- A stable QA build is provided before execution.
- Defect fixes are supplied without long external delays.
- One approved kiosk resolution and Chromium-based browser are used.

Dependencies

- Working Spring Boot backend and test environment.
- Stable UPI and Card payment sandbox.
- Test menu data containing available and unavailable items.
- Valid, invalid, expired, and inapplicable coupon data.
- Test email inbox for receipt delivery.
- Approved test UPI IDs and Card test credentials.
- Jira or an equivalent defect-management tool.

5. WBS Estimation Matrix

WBS ID	QA activity	Main output	Effort (hours)
WBS-01	FRS review and requirement analysis	Clarification list and coverage notes	4
WBS-02	Test planning and estimation	Approved test approach and estimate	4
WBS-03	Test scenario, test case, and test-data design	Review-ready test suite	20
WBS-04	Test review and rework	Approved scenarios and test cases	4
WBS-05	Environment validation and smoke testing	Test-ready build confirmation	4
WBS-06	Food Catalog execution	Catalog results and evidence	3
WBS-07	Cart execution	Cart results and evidence	4

WBS ID	QA activity	Main output	Effort (hours)
WBS-08	Checkout and coupon execution	Checkout results and evidence	6
WBS-09	UPI and Card payment execution	Gateway integration results and evidence	8
WBS-10	Confirmation, receipt, and session execution	Receipt and session results	4
WBS-11	UI, accessibility, privacy, security, and reliability checks	Non-functional results	5
WBS-12	Defect investigation, reporting, and triage	Logged and tracked defects	4
WBS-13	Defect retesting and regression	Fix validation and regression results	10
WBS-14	RTM, execution report, and test closure	Final QA reports	4
Base effort			84
Contingency - 15%		Gateway, defect, and minor clarification uncertainty	13
Total estimated effort			97 hours

6. Effort and Duration Summary

Measure	Estimate
Base QA effort	84 hours
Contingency	13 hours
Total QA effort	97 hours
Productive hours per person-day	7 hours
Estimated effort in person-days	Approximately 14 person-days
Planned duration with 1 QA Engineer	Approximately 14 working days

The duration assumes prompt environment access and defect turnaround. Waiting time for builds, approvals, or external services is not included in QA effort and may extend calendar duration.

Indicative schedule

Phase	Expected duration
Requirement analysis and planning	1-2 days

Phase	Expected duration
Scenario and test-case design	3-4 days
Review, environment validation, and smoke test	1 day
Test execution and defect reporting	4-5 days
Retesting, regression, RTM, and closure	2-3 days
Total	Approximately 14 working days

7. Resource Requirements

Human resources

Role	Allocation	Responsibility
QA Engineer	1 full-time	Test design, data preparation, execution, defects, regression, and reporting
QA Lead	Part-time	Review, risk decisions, estimate approval, and test closure review
Developer / Integration support	As required	Build support, defect clarification, and fixes

Environment and tools

- One touchscreen kiosk or representative landscape test device.
- Supported Chromium-based kiosk browser.
- QA environment connected to test backend services.
- UPI and Card sandbox access.
- Test email inbox.
- Test-document repository and defect-management tool.

8. Risks and Mitigation

Risk	Possible impact	Mitigation
Payment sandbox is unstable or unavailable	Payment cases become blocked and schedule slips	Confirm sandbox access before execution and prepare approved simulated outcomes where permitted.
Coupon, menu, or email test data is unavailable	Related scenarios cannot be completed	Prepare and verify test data before the test cycle begins.
QA build fails smoke testing	Detailed execution cannot start	Apply entry criteria and return an unstable build for correction.
FRS changes after approval	Existing scenarios and estimates become incomplete	Use change control and re-estimate affected requirements.

Risk	Possible impact	Mitigation
High defect density or slow fix turnaround	Retesting and regression exceed the estimate	Prioritise P0/P1 flows, track blocked cases, and revise the schedule when contingency is consumed.

9. Cost and Change Control

No monetary cost is stated because approved resource rates were not provided.

Estimated QA cost = 97 hours × approved blended QA hourly rate

The estimate shall be reviewed when any of the following changes:

- FRS scope or requirement count.
- Supported kiosk devices, resolutions, or browsers.
- Payment methods or payment provider.
- Required test levels, automation, or non-functional coverage.
- Team size, environment readiness, or delivery schedule.

10. Approval

Stakeholder	Representative	Approval method	Date	Status
Prepared by	Kushagra Sinha (QA Engineer)	Email	25 September 2026	Completed
QA Lead	To be assigned	Email / Jira	-	Pending
Project Manager	To be assigned	Email / Jira	-	Pending
Product / Business Stakeholder	To be assigned	Email / Jira	-	Pending